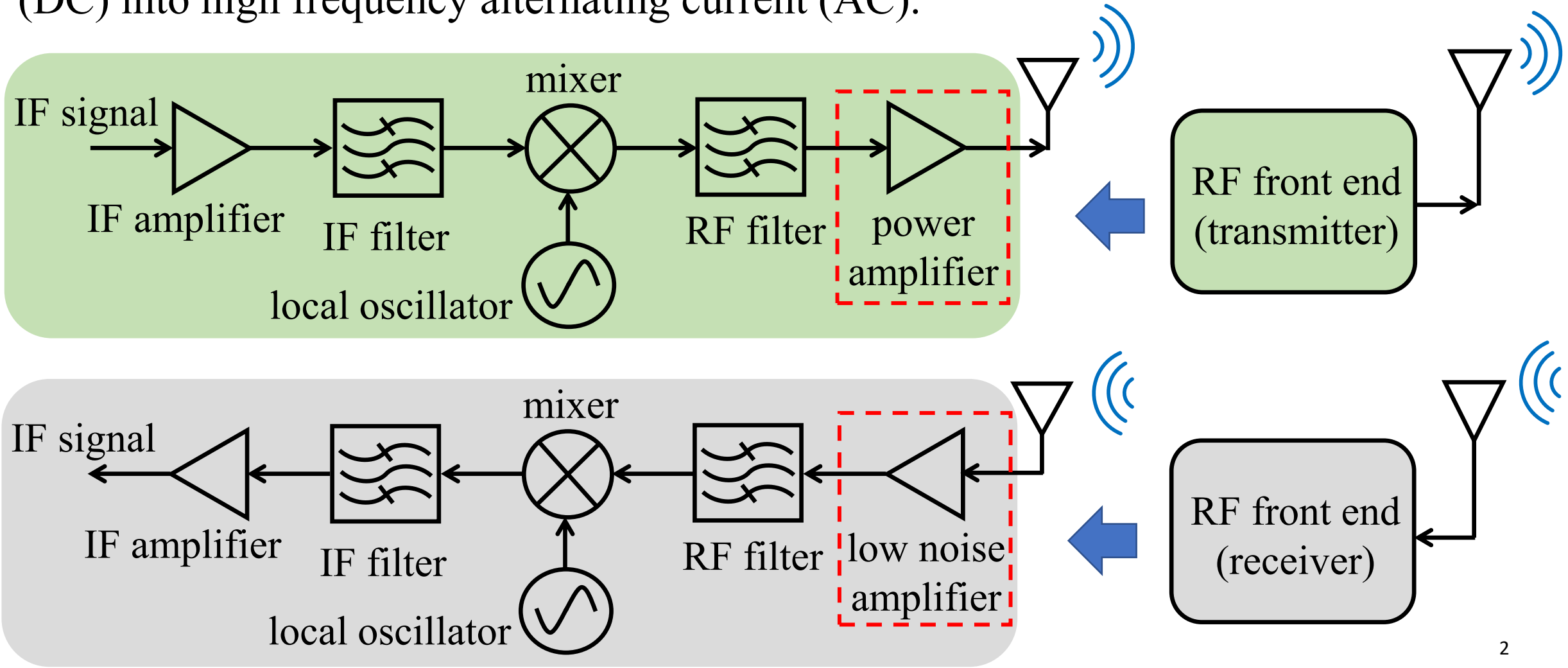
EEME 30004
Communications & Networks

RF components #2 & system analysis

Dr Sean Gu

RF amplifier

RF amplifiers increase the power level of an RF signal by converting direct current (DC) into high frequency alternating current (AC).

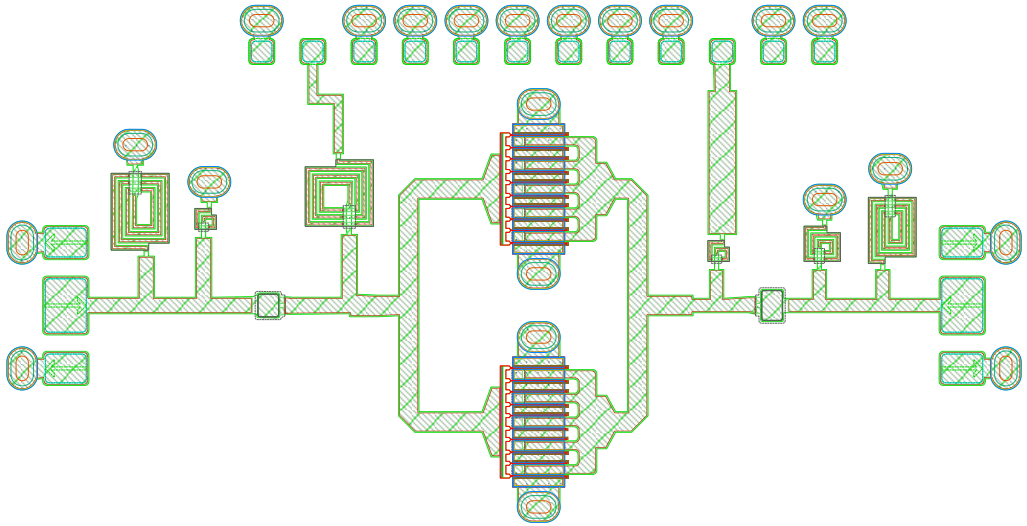


RF amplifier

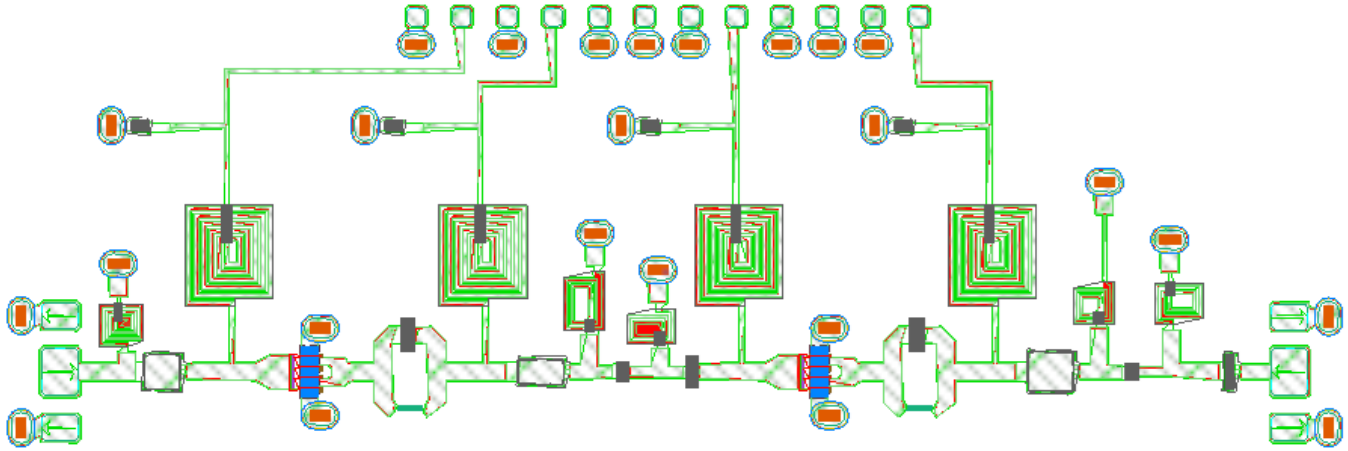
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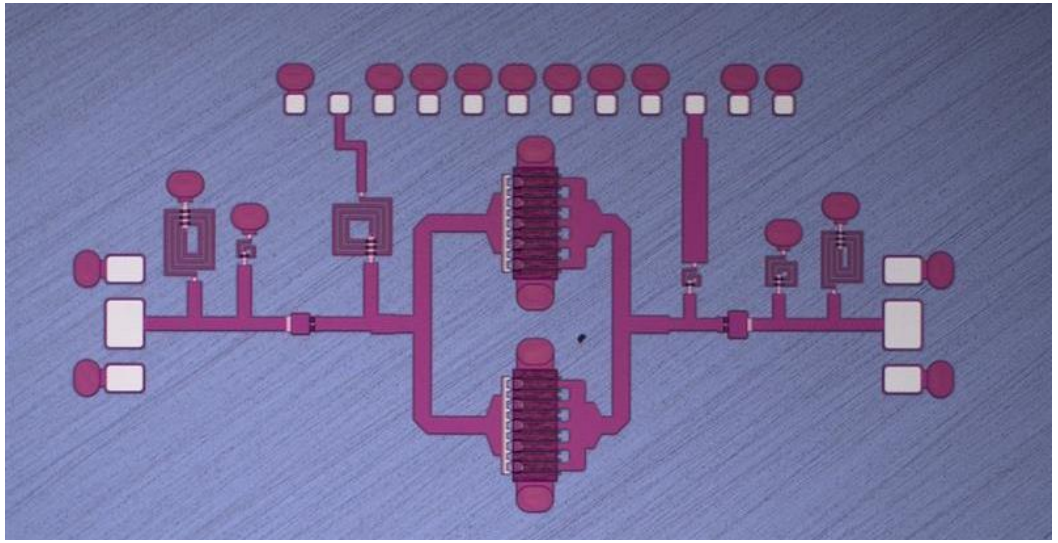
MMIC (monolithic microwave integrated circuit)

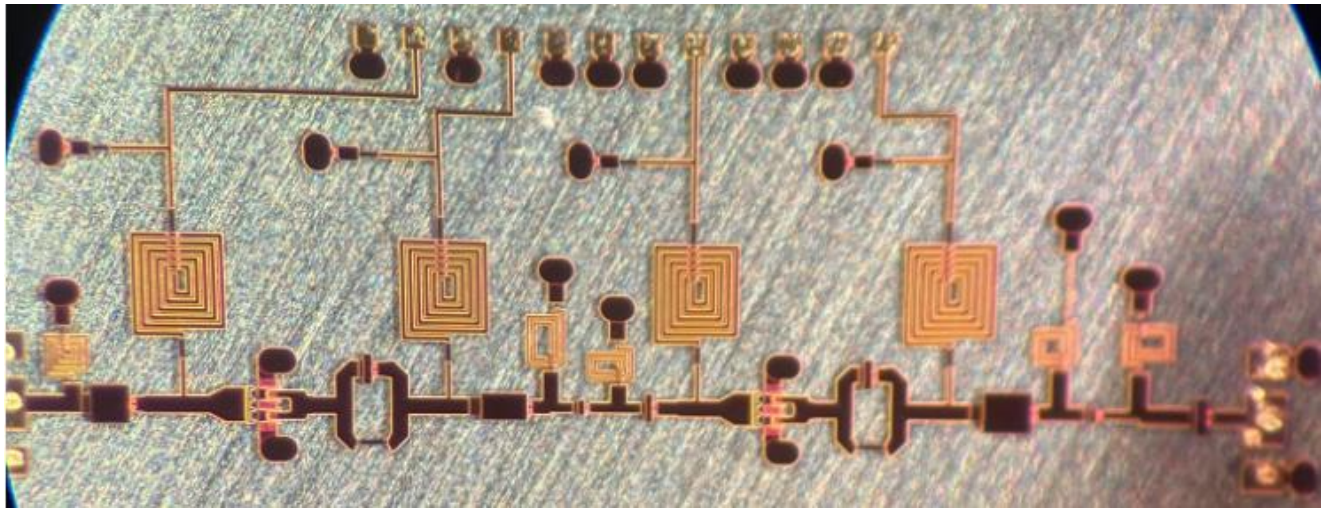
power amplifier (PA)



low noise amplifier (LNA)







RF amplifier

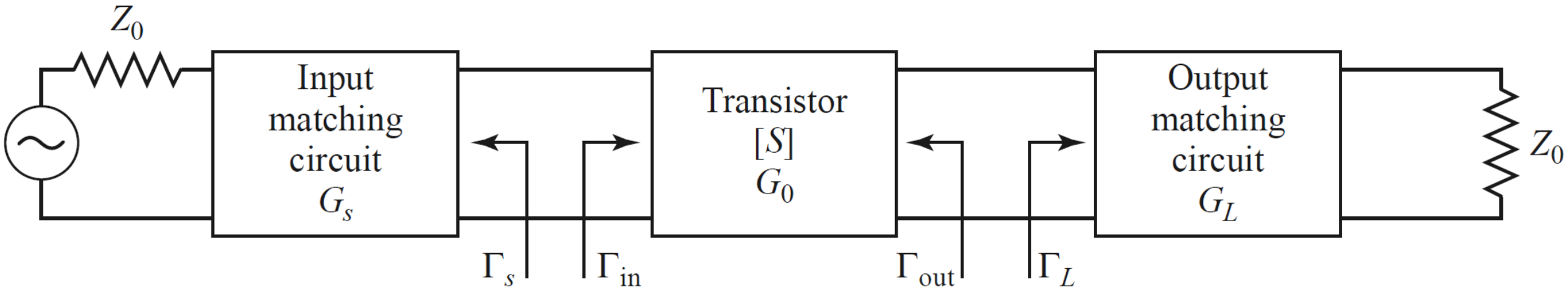
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Specification	Units	Meaning / Why It Matters
Frequency Range	GHz	Frequency band over which amplifier maintains rated gain.
Gain	dB	Ratio of output to input power; defines amplification strength.
Noise Figure (NF)	dB	Measures noise added; low NF is essential for LNAs.
Output Power (Psat, P1dB)	dBm	Maximum power before compression or saturation.
Input/Output VSWR or Return Loss	Ratio / dB	Impedance match quality; affects reflections and efficiency.
Isolation / Reverse Gain	dB	How well amplifier prevents feedback from output to input.
Power Supply	V, mA	Required DC bias for operation.
Connector Type	—	Mechanical interface (e.g., SMA, N-type, 2.92 mm).
Operating Temperature Range	°C	Environmental reliability for test or field use.

RF amplifier

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Amplifier stability is an amplifier's ability to resist self-oscillation and maintain its intended performance without unwanted fluctuations.



① Rollet's condition:

$$\begin{cases} K = \dfrac{1-|S_{11}|^2-|S_{22}|^2+|\Delta|^2}{2\cdot|S_{12}\cdot S_{21}|} > 1 \\ |\Delta| = |S_{11}\cdot S_{22}-S_{12}\cdot S_{21}| < 1 \end{cases}$$

or

② μ test:

$$\mu = \frac{1-|S_{11}|^2}{|S_{22}-\Delta S_{11}^*|+|S_{12}\cdot S_{21}|} > 1$$

RF amplifier

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Example: Design an amplifier using a GaAs FET transistor with following S parameters at 2 GHz:

$$[S]=\begin{bmatrix} 0.64/-170 & 0.05/16 \\ 2.05/27 & 0.56/-96 \end{bmatrix}$$

Stability study: $\Delta=S_{11}S_{22} - S_{12}S_{21} \approx 0.304 \angle 109.2°$

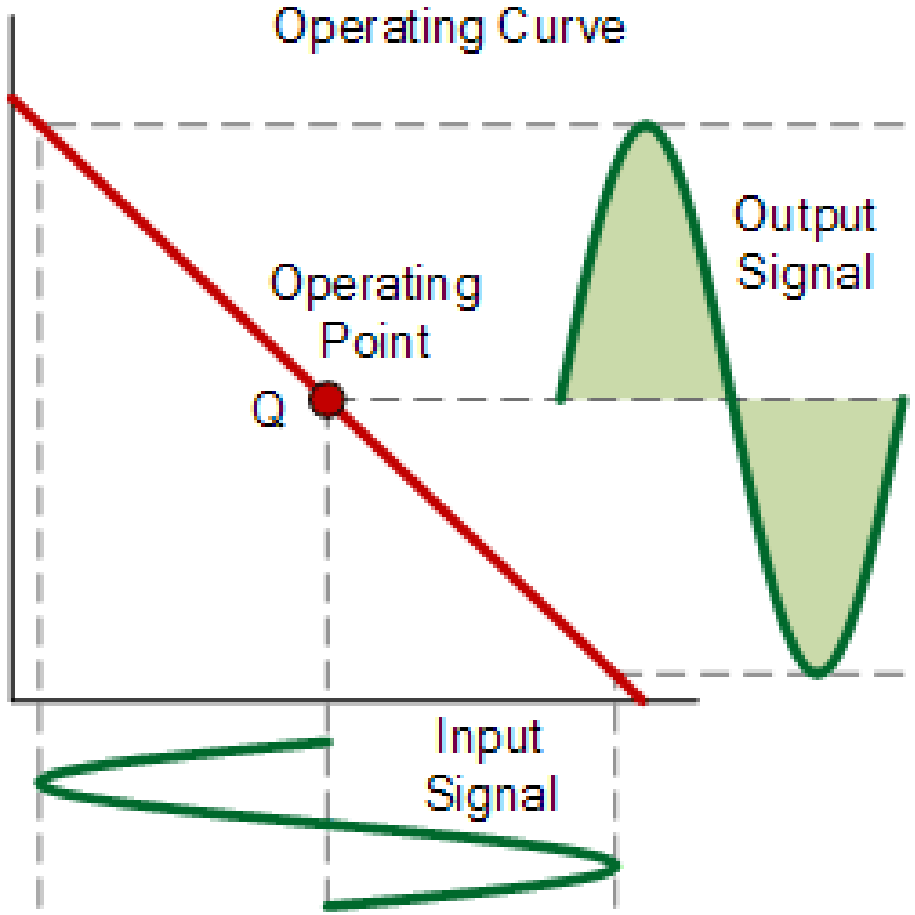
$$K = \frac{1-|S_{11}|-|S_{22}|^2+|\Delta|^2}{2|S_{12}S_{21}|} \approx 1.8$$

We have K>1 and $|\Delta|<1$, the transistor is therefore unconditionally stable.

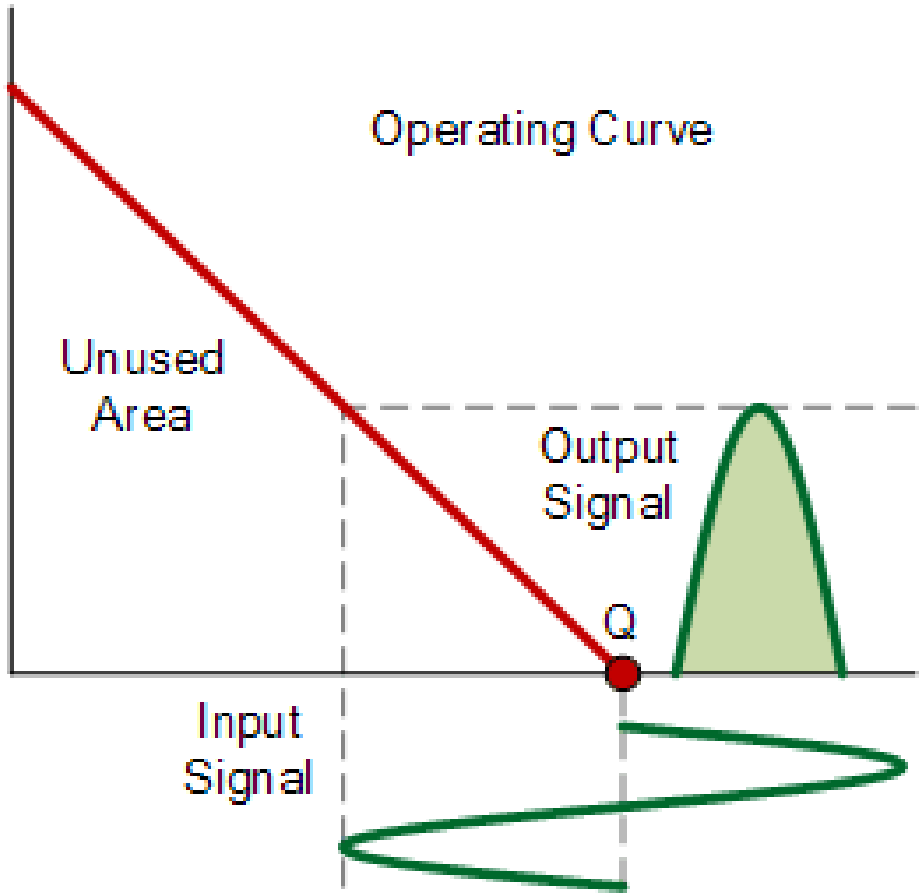
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Class A



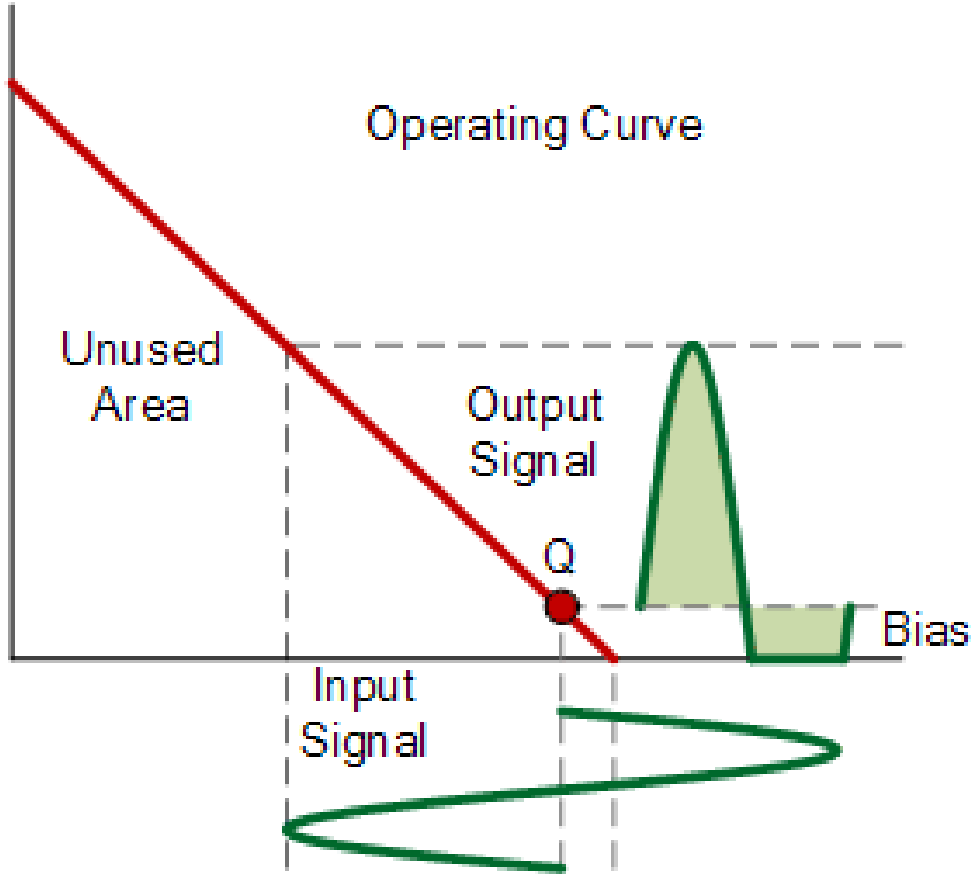
Class B



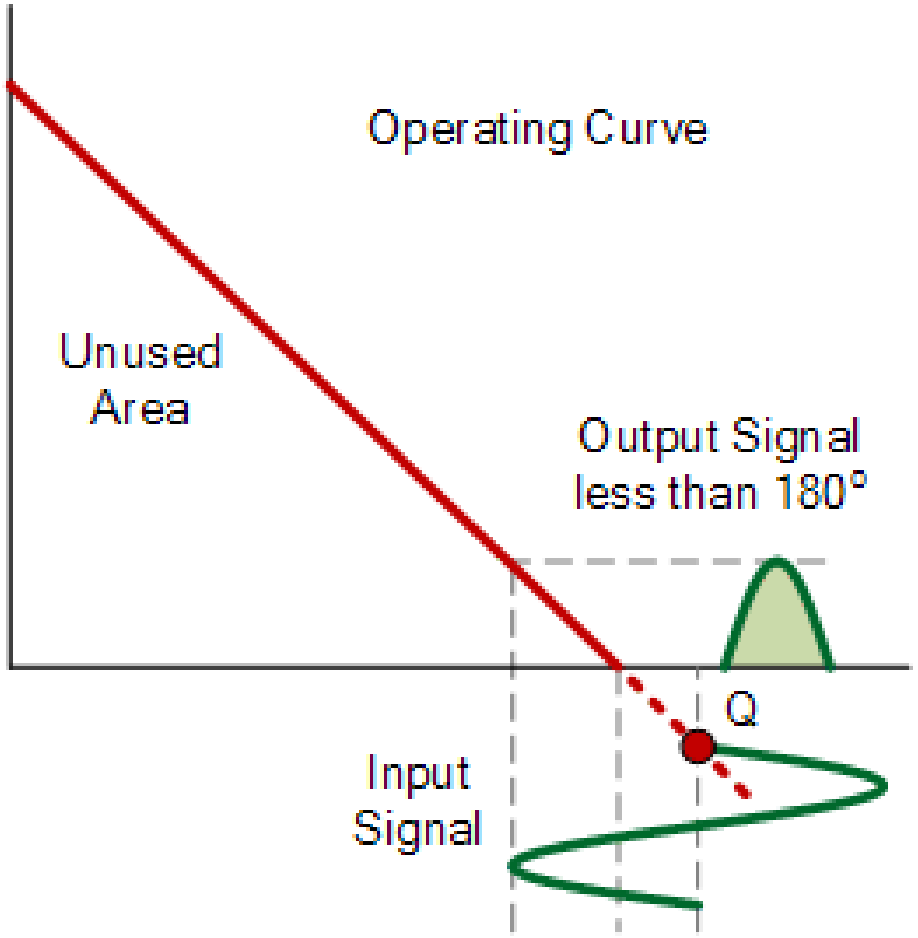
RF amplifier

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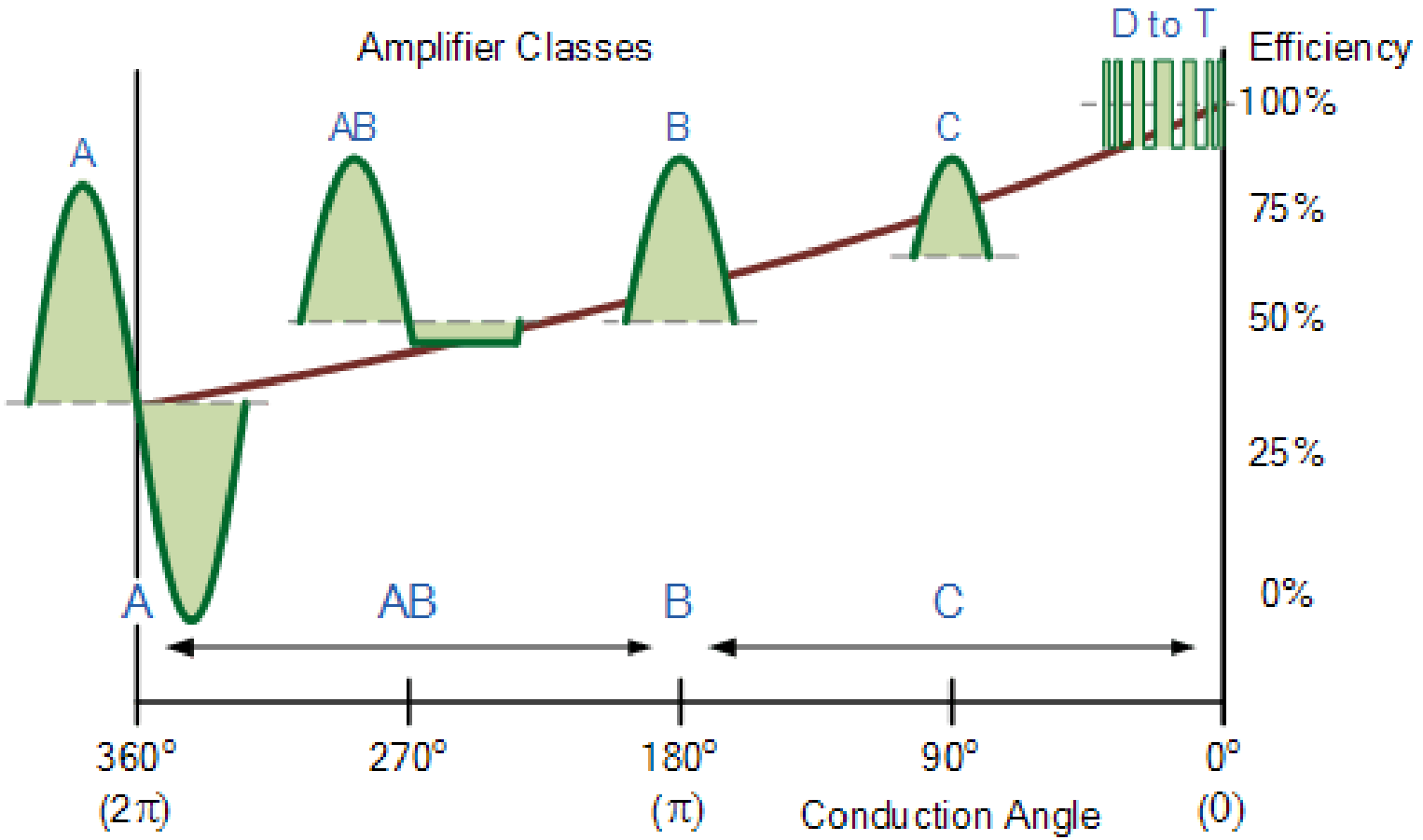
Class AB



Class C



RF amplifier

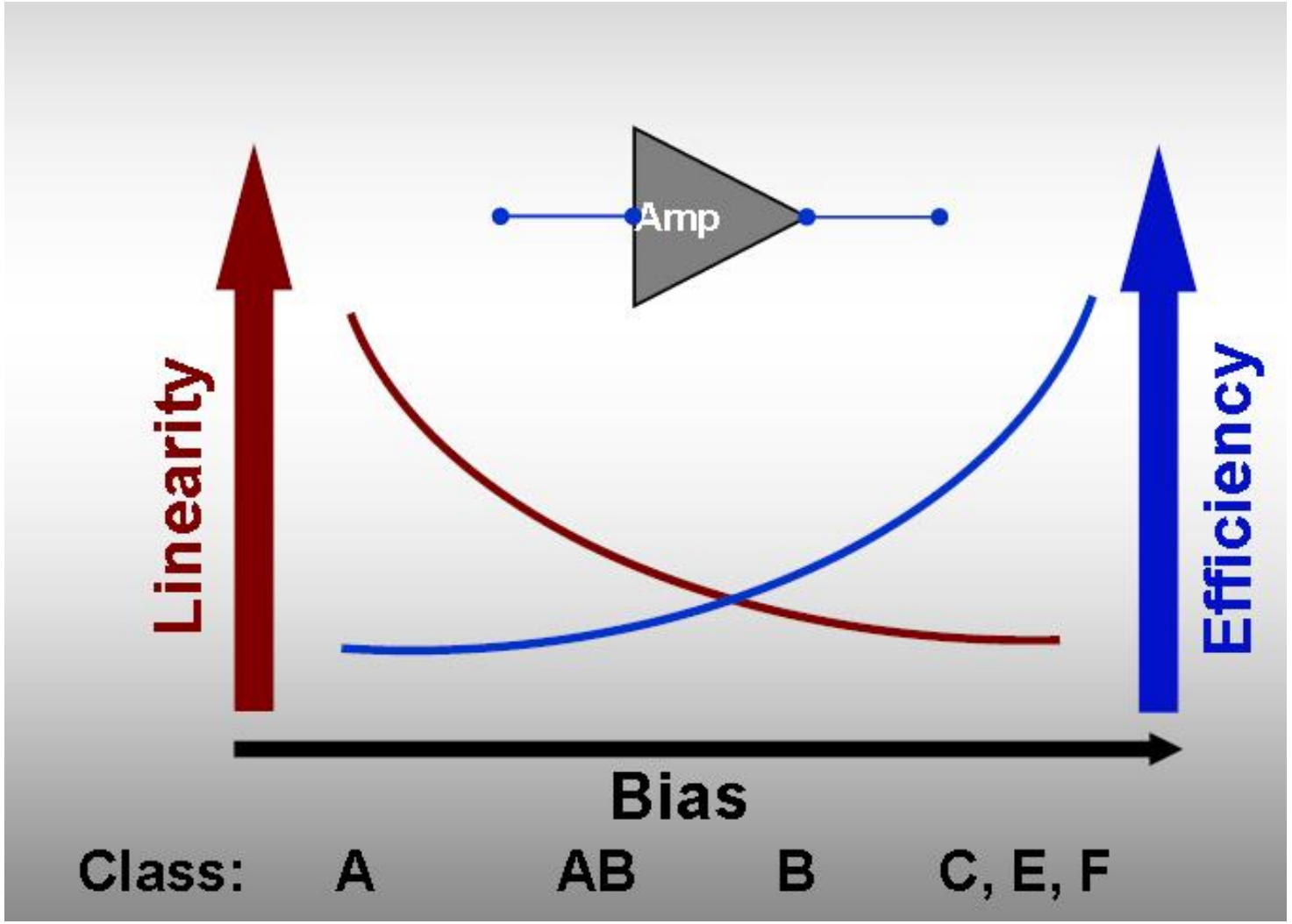


POWER AMPLIFIER CLASSES

CLASS A	CLASS B
• Conduction Angle: 360° • Efficiency: ≈25-30% • Applications: Low-distortion	• Conduction Angle: 180° • Efficiency: 60-70% • Applications: Push-pull
CLASS AB	**CLASS C**
• Conduction Angle: 180° ~ 360° • Efficiency: Up to 80% • Applications: RF transmitters	• Conduction Angle: < 180° • Efficiency: >85% • Applications: UHF transmitters
CLASS E	**CLASS F**
• Operation: Switching: Mode • Efficiency: >85% • Applications: UHF transmitters	• Harmonic Shaping Operation: >85% • Efficiency: >HF-frequency amplifiers

RF amplifier

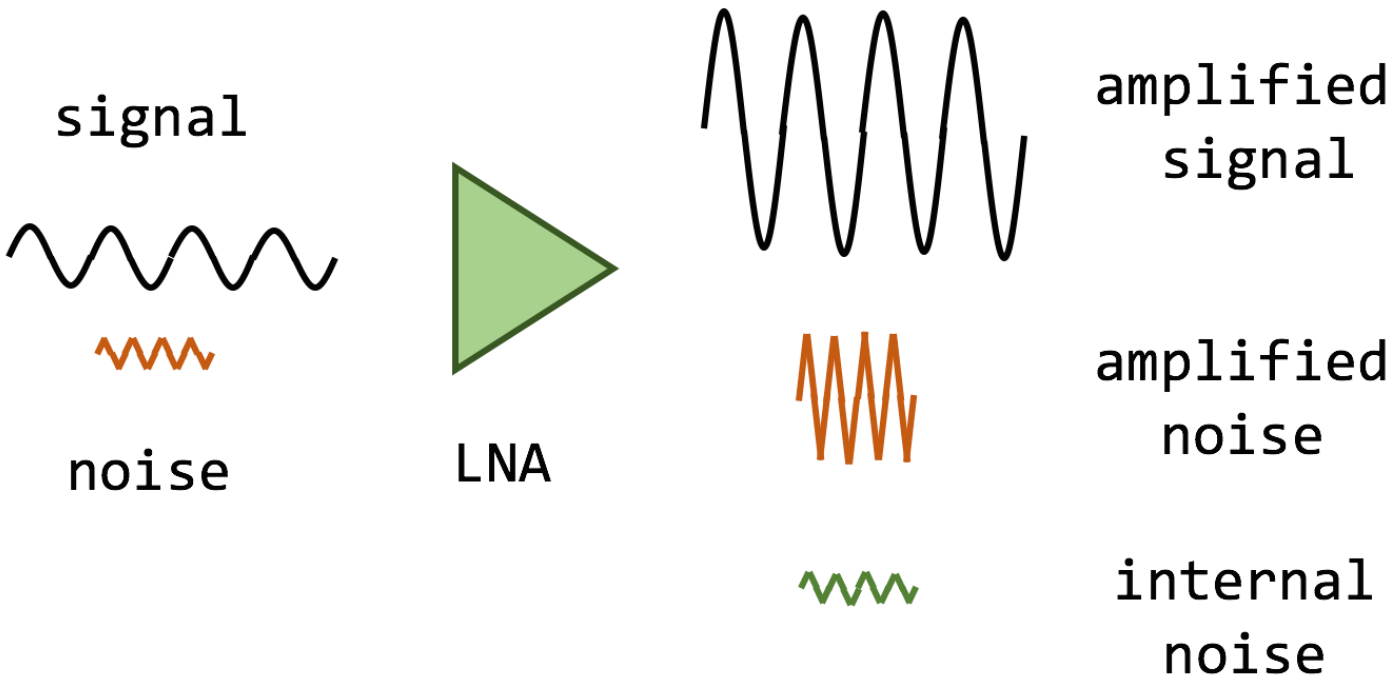
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RF amplifier

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Low-noise amplifiers (LNA) are a class of high gain amplifiers with low noise, good linearity, stable impedance, and low power consumption.



$$NF = \frac{SNR_{in}}{SNR_{out}}$$

ideal case:

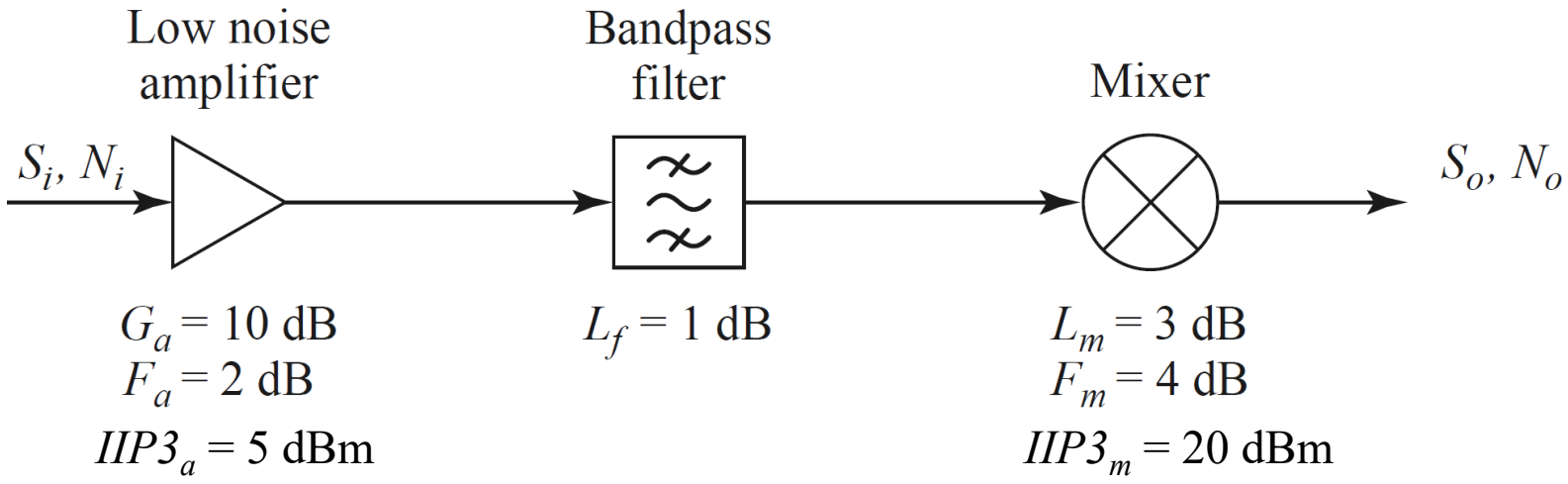
$$SNR_{out} = \frac{P_{signal} \times G}{P_{noise} \times G} = SNR_{in}$$

$$SNR_{out} = \frac{P_{signal} \times G}{P_{noise} \times G + int.noise}$$

LNA: minimize internal noise

System analysis

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1. Total gain (dB) of this cascaded system?
2. IIP3 (dBm) of this cascaded system?
3. NF (dB) of this cascaded system?

$$G_{cas} = G_a + G_f + G_m = 10 + (-1) + (-3) = 6 \text{ dB}$$



Convert to linear scale:

$G_a = 10 \text{ dB} = 10$ $\quad G_f = -1.0 \text{ dB} = 0.79$ $\quad G_m = -3.0 \text{ dB} = 0.5$

$F_a = 2 \text{ dB} = 1.58$ $\quad F_f = 1 \text{ dB} = 1.26$ $\quad F_m = 4 \text{ dB} = 2.51$

$IIP3_a = 5 \text{ dBm} = 3.2 \text{ mW}$ $\quad IIP3_f = ??$ $\quad IIP3_m = 20 \text{ dBm} = 100 \text{ mW}$

Questions?

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